Size Confounds Semantics: An Audit of SING's Null-Space Leakage Score

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Abstract

SING [10] ranks classifiers by how much class-relevant semantics leak into the linear head's null space, and reports that a "DinoViT" pretrained with DINO leaks least. We audit that claim. First, every public ViT-L/16 checkpoint consistent with SING's code or wrappers is supervised or SWAG-supervised; none matches the reported 84.0% top-1, and none is DINO. Second, within a fixed model, SING's absolute-shift score |AS| grows by a factor of ≈ 10 as the removed null subspace grows from d=24 to d=1048 on 3 of 4 models; at matched d=24 the ranking already disagrees with SING's published order (Kendall τ=0.29). A pooled log-linear fit finds $|\mathrm{AS}| \propto \rho^{0.43}$ in removed feature energy ρ, close to a simple $\sqrt{\rho}$ geometric prediction, but the energy-vs-dimension separation is mixed rather than decisive. We define a size-fair score $\mathrm{AS}_{\mathrm{fair}}$ that divides null removal by an energy-matched random-subspace baseline. On all 13 SING models, $\mathrm{AS}_{\mathrm{SING}}$ reproduces the published Fig. 6b ordering (τ=0.82), while $\mathrm{AS}_{\mathrm{fair}}$ breaks it (τ=−0.18); two class-level claims reverse with bootstrap CIs. Planted-leakage recovery is weak for both scores. We recommend reporting ρ and $\mathrm{AS}_{\mathrm{fair}}$ alongside raw AS whenever null-space scores are used to compare architectures.

1 Introduction

A linear classifier on features $f \in \mathbb{R}^m$ has a null space of dimension $m-c$ (for c classes) that does not change the logits. SING [10] argues that class-relevant content in that null space is "leakage": the head could have used it, but did not. Their Absolute Shift (AS) measures how much a CLIP-aligned [5] translation of f moves when the null component is removed, and they use AS (and the IS/AS ratio) to rank 13 ImageNet models, highlighting a ViT labeled "DinoViT" as least leaky [1].

Null-space analyses are also used for OOD detection [2, 9] and overfitting scores [6]. Across architectures, however, both the null dimension $d=m-\mathrm{rank}(W)$ and the null energy fraction $\rho = \mathbb{E}\|V_n V_n^\top f\|^2 / \mathbb{E}\|f\|^2$ vary by orders of magnitude. If AS largely tracks how much energy is removed, cross-model comparisons of "semantics" are confounded by size.

Contributions. (1) We show that SING's "DinoViT" is not identifiable as DINO: every matching candidate is supervised, and none reproduces 84.0% top-1. (2) Within-model AS(d) curves give a $\approx 10\times$ rise from d=24 to d=1048 on ResNet50, ResNeXt101 and DINOv2-L; BiT [3] is an exception explained by translator cancellation. (3) Energy and dimension are hard to separate: a pooled fit yields exponent a=0.43 on ρ vs. a $\sqrt{\rho}$ prediction of 0.5, but fixed-ρ flatness is not clean. (4) On 13 models, size-fair $\mathrm{AS}_{\mathrm{fair}}$ destroys SING's published ranking (τ: 0.82 → −0.18) and reverses several claims, with stated caveats. We propose $\mathrm{AS}_{\mathrm{fair}}$ as a reporting correction, not as a new ground-truth leakage oracle.

2 Background and the DinoViT identity

SING AS in brief. Let $T(f) = Af + b$ be SING's released linear translator into CLIP space and V_n an orthonormal basis of $\ker(W)$ (tolerance 10^{-5}). For a text direction t, SING's per-image absolute shift is the change in angle $\theta(T(f), t)$ after removing $V_n V_n^\top f$. We write $\mathrm{AS}_{\mathrm{SING}}$ for the image-averaged |AS| under full null removal (class text unless noted).

Exp 0b: "DinoViT" is not DINO. SING reports DinoViT at 84.0% ImageNet top-1 (Supp. Table 2). The released models.yaml loads a timm AugReg ViT-L/16; the supplement's wrapper uses torchvision ViT attribute names. We evaluated five public ViT-L/16 candidates on the full 50k validation set (native and force-224): torchvision IMAGENET1K_V1 (79.67%), timm AugReg (85.85%), SWAG linear [7] (85.14%), SWAG E2E (88.07%), and HF google/vit-large-patch16-224 (81.98%). None lies in 84.0 ± 0.3 (Fig. 1). Every candidate is supervised or SWAG weakly supervised; DINO never released a ViT-L with a 1000-way head [1]. The headline attribution to self-supervised DINO therefore does not hold for any identifiable checkpoint.

Exp 0 (re-analysis). Digitizing SING Supp. Fig. 6b IS/AS bars and correlating against measured null dimension gives Spearman −0.46 (p=0.12) over 13 models (DenseNet null dim = 27 at tol 10^{-5}). Null energy alone is more predictive: $\mathrm{Spearman}(\rho_{\mathrm{full}}, \mathrm{IS/AS}) = -0.94$ (p=1.9×10^{-6}). DeiT3-L [8] has null dim 24 yet the worst ratio—dimension alone cannot explain the ranking, which motivates a within-model test.

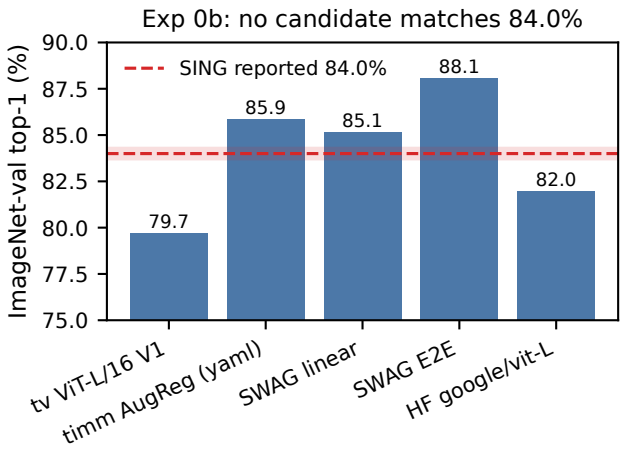


Figure 1: Exp 0b top-1 on ImageNet-val. No candidate matches SING's reported 84.0%.

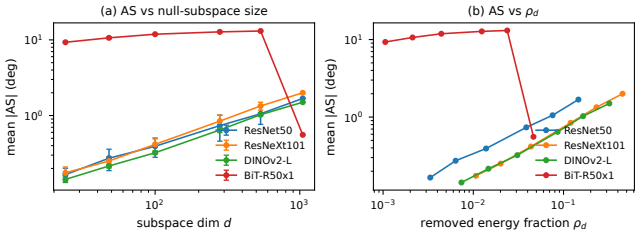


Figure 2: Exp 1: mean |AS| vs. subspace size d (a) and removed energy ρ_d (b). BiT collapses at the full null (translator cancellation).

3 Within-model size dependence (Exp 1)

On a fixed 10k ImageNet-val subset (10 images/class), we remove random d-dimensional subspaces of each model's own null space ($d \in \{24, 48, 100, 280, 536, 1048\}$, 20 seeds for d<d_{null}) for ResNet50, ResNeXt101, BiT-R50x1 and DINOv2-L (all d_{null}=1048), using SING's released translators.

Fig. 2 and Table 1 show r=AS(1048)/AS(24) ≈ 10 on 3 of 4 models. At matched d=24 over 8 models, Kendall τ vs. the Fig. 6b order is 0.29 (p=0.40) for |AS|. Both pre-registered GO conditions hold.

BiT anomaly. BiT's full-null AS collapses (r=0.06) while partial subspaces give large AS (∼9–13°). Diagnostics show extreme singular values of AV_n (top 284 vs. 5.6 for ResNet50): orthogonal null blocks nearly cancel through the translator when summed, so full-null AS understates BiT's per-direction sensitivity. This is translator conditioning, not a code bug.

Caveat. For random subspaces, ρ_d is nearly linear in d, so R^2 vs. d and vs. ρ_d are identical (≈ 0.98). Exp 1 alone cannot tell energy from dimension.

Model	r	2σ band	R^2 vs d	R^2 vs ρ_d
ResNet50	10.15	[5.83, 14.48]	0.983	0.983
ResNeXt101	11.44	[6.93, 15.95]	0.977	0.977
DINOv2-L	10.48	[8.85, 12.10]	0.974	0.974
BiT-R50x1	0.060	[0.057, 0.063]	0.512	0.512

Table 1: Exp 1: r=AS(1048)/AS(24). Energy and dimension fits are collinear for random subspaces.

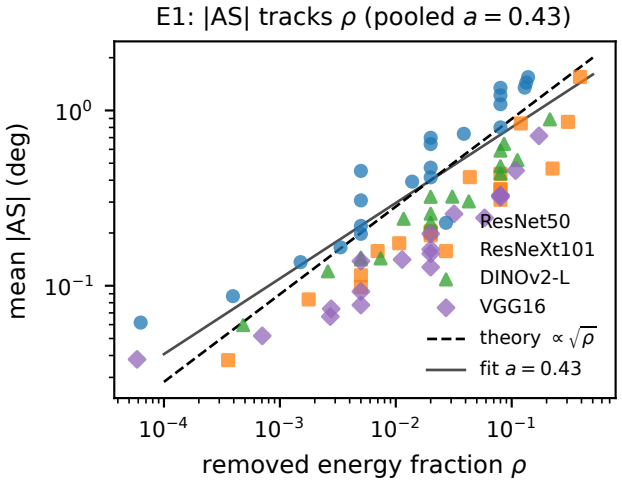


Figure 3: E1 cells: |AS| vs. ρ. Dashed: $\sqrt{\rho}$ theory; solid: fitted a=0.43.

4 Energy vs. dimension (E1): a mixed result

Geometry. If a linear translator maps a removed feature component of relative energy ρ into CLIP space, the first-order angular displacement scales with the removed magnitude $\|\Pi f\| \propto \sqrt{\rho}$ when the residual is small relative to $\|T(f)\|$. This predicts $|\text{AS}| \propto \rho^{1/2}$ at fixed geometry—an exponent of 0.5.

Design. On ResNet50, ResNeXt101, DINOv2-L, VGG16 and EVA02-L we (a) compare high-energy vs. low-energy null PCs at fixed $d \in \{24, 100, 280\}$ and (b) hit fixed $\rho \in \{0.5\%, 2\%, 8\%\}$ at several d by mixing top/bottom PCs. Pooled fit over 76 feasible cells:

$$\log|\text{AS}| \;=\; a\log\rho \;+\; b\log d \;+\; c_{\text{model}},$$

gives a=0.431 [0.430, 0.433], b=0.010 [0.008, 0.012], R^2=0.93 (Fig. 3).

High/low energy ratios at fixed d exceed 3× on all 4 evaluable models (e.g. ResNet 11–22×, ResNeXt ≈ 10×, DINOv2 3.7–5.1×, VGG16 6–10×). Fixed-ρ max/min across d, however, exceeds 1.5 on several cells (ResNet up to 2.29 at ρ=0.5%). EVA02-L (d_{null}=24) is not evaluable beyond its full-null cell. We therefore call E1 mixed, not decisive: energy dominates the fit and matches $\sqrt{\rho}$ to within $\approx$ 0.07 in the exponent, but the pre-registered ENERGY-DRIVEN flatness rule is not cleanly met.

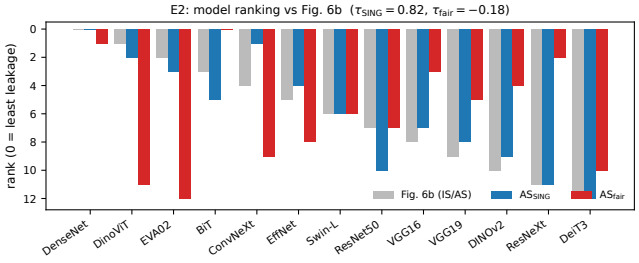


Figure 4: Ranks (0=least leakage). AS_{SING} tracks Fig. 6b; AS_{fair} does not.

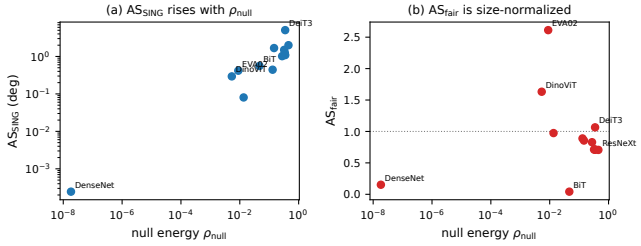


Figure 5: (a) AS_{SING} rises with ρ_{null}. (b) AS_{fair} removes that trend.

ID	Claim (short)	AS_{SING}	AS_{fair}
C1	DinoViT best IS/AS among 5	holds	reverses
C2	ResNeXt lowest IS/AS among 5	holds	reverses
C3	ResNet leaks more than DinoViT	holds	reverses
C4	DinoViT least class leakage	holds	reverses
C5	ResNeXt high AS (top-2 of 5)	holds	reverses
C6	ResNet per-class AS > DinoViT	holds	reverses
C7	DinoViT per-class |AS|<1	holds	reverses
C8	Porcupine/sports-car ≥90th pct.	reverses	reverses
C9	No ResNet–DinoViT class corr.	holds	reverses
C10	Jellyfish ≫ camel (DinoViT)	reverses	reverses
C11	DinoViT least steerable (proxy)	holds	reverses
C12	Fig. 6b 13-model order	holds	reverses
C13	EVA02 target-space order	untestable	untestable

Table 2: E2 claim audit. C1/C2/C11 use |AS| proxies; C10 is class-text on 50 imgs/class.

5 A size-fair score

Define the energy-matched random baseline: draw a Haar subspace of the full feature space that removes exactly the same energy $\mathbb{E}\|\Pi f\|^2$ as the null projector, and let AS_{rand} be its mean |AS| (20 draws). Then

$$AS_{fair} = \frac{AS_{null}}{AS_{rand}}.$$

Values near 1 mean the null is no more shifting than a random subspace of equal energy; values $\gg 1$ mean concentrated null leakage. This is the E3-registered definition, reused at α=0 for the E2 audit.

6 Planted leakage (E3)

We plant a CLIP concept direction (red/striped/water) into each of 8 models' null spaces at log-uniform $\alpha \in [0.01, 1]$ over 50 trials, leaving logits unchanged (max $|\Delta\text{logit}| \leq 2.3\times10^{-5}$). Kendall τ vs. the planted-α ranking, concept-averaged: AS_{SING} 0.386 $[0.324, 0.447]$; AS_{fair} 0.071 $[0.004, 0.138]$. Neither recovers planted strength well; AS_{fair} is near chance. Per-concept means are similar (red 0.386/0.070, striped 0.403/0.061, water 0.369/0.083). Raw observations: DeiT3-L signed AS is negative at every α and |AS| shrinks as α grows; BiT AS_{SING} reaches 17–20° at α=1 while its AS_{fair} at α=0 is only 0.045–0.063; AS_{fair} is non-monotonic in α for EVA02-L and ConvNeXt-B. We report this honestly: size-fairness and planted-ranking recovery are different goals. AS_{fair} is proposed only as a size correction for cross-model AS comparisons, not as a calibrated leakage oracle.

7 Audit of SING's 13-model claims (E2)

Using the same 10k subset plus a 200-image claim-class set (all 50 val images of jellyfish, porcupine, Arabian camel, sports car), we recompute AS_{SING} and AS_{fair} for all 13 released translators.

Ranking. Vs. digitized Fig. 6b IS/AS order: $\tau(AS_{SING})$=0.821 (p=1.3×10^{-5}); $\tau(AS_{fair})$=−0.179 (p=0.435). Without DenseNet: $\tau(AS_{fair})$=−0.364. Mean over 173 SING concepts: τ=0.641 vs. −0.026. Our setup reproduces SING when using AS_{SING}; disagreement appears only after size correction (Figs. 4–5).

Claim reversals. Under AS_{fair}, claims C1–C7, C9, C11, C12 reverse relative to AS_{SING} (DinoViT no longer least leaky; ResNeXt no longer among the worst). Two claims reverse under both scores, with bootstrap CIs:

- C8 (ResNet porcupine / sports-car ≥90th percentile): porcupine percentile 84.8 $[80.1, 87.7]$ (AS_{SING}), 68.4 $[64.5, 71.5]$ (AS_{fair}); sports car 61.0 $[56.1, 66.2]$ and 67.2 $[63.6, 71.4]$.
- C10 (DinoViT jellyfish ≫ camel): jellyfish−camel CI $[-0.205, -0.010]$ (AS_{SING}) and $[-1.403, -0.458]$ (AS_{fair})—the sign is opposite SING's claim.

Caveats (must-state). (i) C1/C2 are IS/AS-ratio claims tested here on |AS|; C11 is a steering claim proxied by null-removal AS; C10 uses class-text AS on 50 images/class (SING used ≈30 concepts / ≈1300 images). (ii) DenseNet has ρ_{null}=1.8×10^{-8}, so AS_{fair} is degenerate. (iii) C13 (EVA02 as target space) is untestable: no EVA02-space translators were released. (iv) Our DinoViT IS/|AS| is 20.95 vs. 70.16 in Fig. 6b—absolute ratios differ, but AS_{SING} ranking still agrees.

Under AS_{fair}, DinoViT ranks near the worst end (12th of 13 by class AS_{fair}; EVA02 worst), whereas under AS_{SING} it ranks 3rd. BiT becomes best under AS_{fair} (0.042) precisely because random subspaces of matched energy move AS by 13.2°—the translator-cancellation regime of Exp 1. DenseNet's near-zero ρ makes AS_{fair}

numerically unstable; we therefore also report τ without it (-0.364).

8 Discussion

SING's within-model AS is a useful probe of translator-mediated null content, but as a cross-model score it conflates semantics with how large and energetic the null is. The boring baseline already hinted at this: over 13 models, Spearman($\rho_{\text{full}}, |\text{AS}|_{\text{full}}$)=0.87 and Spearman($\rho_{\text{full}}, \text{IS/AS}$)=$-0.94$. The DinoViT label is additionally unsupported by any public checkpoint.

What AS_{fair} is and is not. It is a reporting correction: divide null AS by an energy-matched random baseline so that models are not rewarded for having a small or low-energy null. It is not a validated ground-truth leakage measure—E3 shows that neither score recovers planted α rankings at the pre-registered $\tau \geq 0.8$ bar, and AS_{fair} is worse at that particular task. We do not claim any replacement score beyond this size correction.

Recommendations. For any null-space leakage comparison:

1. Name the exact checkpoint and measured top-1 (avoid unverifiable labels such as "DinoViT").
2. Report d_{null} and ρ_{null} next to the score.
3. Report AS_{fair} (or another energy-matched baseline) beside raw AS.
4. Flag translator condition numbers when full-null and block-wise AS disagree (BiT).
5. Prefer within-model interventions (subspace size, planted signal) over cross-architecture league tables when d and ρ vary widely.

Limitations. E1 is mixed rather than decisive on energy vs. dimension. E3 shows AS_{fair} does not recover planted α. Several E2 claims use proxies (C1/C2/C10/C11); DenseNet's AS_{fair} is degenerate; C13 is untestable. ImageNet-derived features restrict redistribution. We did not retrain translators, evaluate a true DINO ViT-B/16 [1], or extend the size audit to ViM [9] / Cook [2] / Rezaei [6] scores [4].

9 Conclusion

SING's AS ranking of 13 classifiers is reproducible with the released translators, but size-confounded: correcting for null energy with AS_{fair} breaks the published order (τ: 0.82 $\rightarrow$ -0.18) and reverses multiple claims, while the "DinoViT" result cannot be attributed to DINO. Until a calibrated leakage oracle exists, cross-model null-space comparisons should report ρ and a size-fair baseline alongside raw AS.

Reproducibility. Analysis code and aggregated tables: https://github.com/sudeshgm/ic014-sing-null-dimension. ImageNet features are private.

References

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A Additional numerical detail

Exp 1 full-null anchors. Mean |AS| at full null: ResNet50 1.682°, ResNeXt101 1.999°, BiT 0.556°, DINOv2-L 1.499°; at d=24: 0.166, 0.175, 9.30, 0.143 (seed SD in RESULTS.md).

E2 per-model class scores. Table 3 lists ρ_{null}, AS_{SING}, AS_{rand} and AS_{fair} used for the ranking.

Model	ρ_{null}	AS_{SING}	AS_{rand}	AS_{fair}
VGG16	0.350	1.088	1.537	0.708
VGG19	0.339	1.256	1.755	0.716
DenseNet121	1.8e-8	0.00025	0.0016	0.152
ResNet50	0.146	1.676	1.960	0.855
DinoViT	0.0054	0.294	0.180	1.632
EffNet-B0	0.130	0.443	0.498	0.889
BiT-R50x1	0.046	0.557	13.24	0.042
ResNeXt101	0.451	1.999	2.829	0.707
ConvNeXt-B	0.0135	0.080	0.083	0.974
Swin-L	0.274	1.019	1.229	0.829
DINOv2-L	0.322	1.499	2.105	0.712
DeiT3-L	0.348	5.048	4.738	1.065
EVA02-L	0.0089	0.417	0.160	2.612

Table 3: E2 class-text scores on the 10k subset (from e2_models.csv).

E3 per-concept τ. AS_{SING} / AS_{fair}: red 0.386 / 0.070; striped 0.403 / 0.061; water 0.369 / 0.083.